

FORTIFY™ ELASTOMER C5070D

POLYOLEFIN ELASTOMER

DESCRIPTION

FORTIFY™ Polyolefin Elastomer (POE) C5070D is an ethylene octene copolymer produced by solution polymerization using metallocene catalyst. This product is available as free flowing pellets.

FORTIFY™ Polyolefin Elastomer (POE) C5070D is designed as a low density and high performance copolymer modifier to provide superior impact properties and flow characteristics.

TYPICAL APPLICATIONS

Impact modifier in thermoplastic olefin compounds, footwear midsoles and wire and cable extrusion.

TYPICAL PROPERTY VALUES

Revision 20211208

PROPERTIES	TYPICAL VALUES	UNITS	TEST METHODS
POLYMER PROPERTIES			
Density	868	kg/m ³	ASTM D792
Melt Flow Rate (MFR)			
at 190°C and 2.16 kg	5.0	g/10 min	ASTM D1238
at 230°C and 2.16 kg	11.0	g/10 min	ASTM D1238
Mooney viscosity			
ML 1+4, 121 °C	8	MU	ASTM D1646
MECHANICAL PROPERTIES ⁽¹⁾			
Tensile Properties			
strength at break	6	MPa	ASTM D638
elongation	1100	%	ASTM D638
100% modulus	2.3	MPa	ASTM D638
Durometer Hardness			
shore A (1 second)	63	-	ASTM D2240
shore D (1 second)	16	-	ASTM D2240
Flexural Modulus (1% Secant)	10.8	MPa	ASTM D790 A
Tear Strength (Type C)	35.3	kN/m	ASTM D624
THERMAL PROPERTIES			
Peak Melting Temperature	62	°C	SABIC method
Glass Transition Temperature, T _g	-52	°C	SABIC method

(1) All physical properties were measured from specimens cut from compression molded. These typical values depend on manufacturing conditions. Therefore, customers should confirm the product performance by using their own tests.

STORAGE AND HANDLING

POE Polyolefin Elastomer resins (in pelletized form) should be stored in such a way that it prevents exposure to direct sunlight and/or heat, as this may lead to quality deterioration. The storage location should also be dry, dust free and the ambient temperature should not exceed 30°C. Further avoid temperatures above 50°C and below 10°C. Please mind the temperature conditions when using the low density grades <0.875 g/cm³, especially when the shipment or storage temperature would approach the softening and melting temperature of the POE resin. Outer package wrap should not be removed from the pallets until the products are ready to be used. Stacking of pallets is not recommended due to dimensional instability and material blocking risk. Grades with D suffix are being treated with anti-caking dust agent to reduce blocking behaviour. It is advisable to process Polyolefin Elastomers resins within 6 months after delivery, this because also excessive aging can lead to a deterioration in quality.



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